2D Arrays

& 2D ArrayLists

Contents

Representation of
2D Array / Matrix

Grid

	0	1	2	3	4
0	2	8	3	4	7
1	7	2	1	6	3
2	5	5	4	1	4
3	3	1	8	2	6

arr 4×5

col

arr[1][2]

row

Representation of
2D Array / Matrix

```
int[][] arr = new int[3][2];
```

$$\begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 0 & 0 \end{array}$$

Representation of
2D Array / Matrix

array of arrays

$$\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 2 & 9 & 5 \\ 1 & 8 & 1 & 6 \\ 2 & 3 & 7 & 4 \end{array}$$

arr

$$arr = \{\underset{0}{\{2, 9, 5\}}, \underset{1}{\{8, 1, 6\}}, \underset{2}{\{3, 7, 4\}}\}$$

arr[2][1]

Input & Output in
2D Arrays

Ques: Sum of elements in given Matrix

HW: Find the maximum element in given 2D array.

Ques: Find the row with maximum sum

	0	1	2	3	4	
0	2	8	3	4	7	24
1	7	2	1	6	3	19
2	5	5	4	1	4	19
3	3	1	8	2	6	20

arr 4×5

HW: **Find the minimum element out of all the maximum elements of each row**

	0	1	2	3	4
0	2	8	3	4	7
1	7	2	1	6	3
2	5	5	4	1	4
3	3	1	8	2	6

arr 4×5

Ques: Print elements of 2D Array column-wise

	0	1	2	3	4
0	2	8	3	4	7
1	7	2	1	6	3
2	5	5	4	1	4
3	3	1	8	2	6

arr 4×5

2	7	5	3
8	2	5	1
3	1	4	8
4	6	1	2
7	3	4	6

0^{th} col → arr[0][0] arr[1][0] arr[2][0] arr[3][0]

Ques: Print matrix in Snake Pattern

$$
\begin{array}{c|ccccc}
 & 0 & 1 & 2 & 3 & 4 \\
\hline
0 & 2 & 8 & 3 & 4 & 7 \\
1 & 7 & 2 & 1 & 6 & 3 \\
2 & 5 & 5 & 4 & 1 & 4 \\
3 & 3 & 1 & 8 & 2 & 6
\end{array}
$$

arr 4×5

Output :

2 8 3 4 7

3 6 1 2 7

5 5 4 1 4

6 2 8 1 3

Output : 2 8 3 4 7 3 6 1 2 7 5 5 4 1 4 6 2 8 1 3

HW: Reverse all rows of a given matrix & then reverse the col's.

HW: Snake print column wise

	0	1	2	3	4
0	2	8	3	4	7
1	7	2	1	6	3
2	5	5	4	1	4
3	3	1	8	2	6

arr 4×5

Ques: Transpose of Matrix

$$
\begin{array}{c|cccc}
 & 0 & 1 & 2 & 3 \\
\hline
0 & 2 & 8 & 3 & 4 \\
1 & 7 & 2 & 1 & 6 \\
2 & 5 & 5 & 4 & 1 \\
3 & 3 & 1 & 8 & 2
\end{array}
\quad \Rightarrow \quad
\begin{array}{c|cccc}
 & 0 & 1 & 2 & 3 \\
\hline
0 & 2 & 7 & 5 & 3 \\
1 & 8 & 2 & 5 & 1 \\
2 & 3 & 1 & 4 & 8 \\
3 & 4 & 6 & 1 & 2
\end{array}
$$

arr $\qquad\qquad\qquad$ arr

Ques: Transpose of Matrix (Galti)

	0	1	2	3
0	~~2~~ 2	8 ~~8~~ ~~7~~	~~3~~ 3 ~~5~~	4 ~~4~~ ~~8~~
1	7 ~~7~~ ~~8~~	~~2~~ 2	1 ~~1~~ ~~5~~	~~6~~ 6 ~~1~~
2	5 ~~5~~ ~~3~~	5 ~~5~~ ~~1~~	~~4~~ 4	1 ~~1~~ ~~8~~
3	3 ~~3~~ ~~4~~	1 ~~1~~ ~~6~~	8 ~~8~~ ~~1~~	~~2~~ 2

arr

wrong

	0	1	2	3
0	2	~~8~~ 7	~~3~~ 5	~~1~~ 3
1	~~7~~ 8	2	~~1~~ 5	~~6~~ 1
2	~~8~~ 3	~~5~~ 1	4	~~1~~ 8
3	~~3~~ 4	~~1~~ 6	1 ~~8~~	2

arr

Correct

Ques: Transpose of Matrix (In a new matrix)

$$
\begin{array}{c|cccc|}
 & 0 & 1 & 2 & 3 \\
\hline
0 & 1 & 6 & 8 & 3 \\
1 & 4 & 2 & 7 & 5 \\
\hline
\end{array}
$$

a

$$
\begin{array}{c|cc|}
 & 0 & 1 \\
\hline
0 & 1 & 4 \\
1 & 6 & 2 \\
2 & 8 & 7 \\
3 & 3 & 5 \\
\hline
\end{array}
$$

b

$$b[i][j] = a[j][i]$$

Ques: Rotate by 90 degree

2	8	3	4
7	2	1	6
5	5	4	1
3	1	8	2

arr

transpose

2	7	5	3
8	2	5	1
3	1	4	8
4	6	1	2

3	5	7	2
1	5	2	8
8	4	1	3
2	1	6	4

arr

2D ArrayLists & variable sized arrays in 2D array.

11:45 resume

Arraylist of Arraylists

2D ArrayLists

$$\begin{array}{ccccccc} & & 0 & 1 & 2 & 3 & 4 \\ arr & = & \{10, & 20, & 30, & 80, & 110\} \end{array}$$

```
sout(arr.get(2))
arr.set(2, 300)
```

$$\begin{array}{ccccc} & & 0 & 1 & 2 \\ arr & = & \{\{1, 2, 3\}, & \{7, 9, 8, 1\}, & \{4, 5, 3\}, \\ & & \{3\} & & \end{array}$$

```
sout(arr.get(1).get(3))
arr.get(0).set(1, 7)
arr.get(2).add(3)
arr.add(new ArrayList<>());
```

Ques: Pascal Triangle

$n = 5$

	0	1	2	3	4
0	1				
1	1	1			
2	1	2	1		
3	1	3	3	1	
4	1	4	6	4	1

arr

$$arr[i][j] = arr[i-1][j] + arr[i-1][j-1]$$

Ques: Search in a Row-Column Sorted matrix

1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	14	17	24
18	21	23	26	30

tar = 17

1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	14	17	24
18	21	23	26	30

target = 20

Ques: Search in a Row-Column Sorted matrix

1	4	7	11	15
2	5	8	12	19
2.5	6	9	16	22
10	13	14	17	24
18	21	23	26	30

target = 3

```
int i = 0, j = n-1
while ( j>=0 && i< m ){
    if(arr[i][j] > target) j--
    else if(arr[i][j] <tar) i++
    else return true
}
return false
```

Ques: Spirally Traversing a Matrix

	fc					lc
fr	1	2	3	4	5	6
	7	8	9	10	11	12
	13	14	15	16	17	18
	19	20	21	22	23	24
lr	25	26	27	28	29	30

m x n

Right → first row st.col to end col

Down → last col up to down

left → last row r to l

Up → first col d to u

Ques: Spirally Traversing a Matrix

fc lc

lr

fr

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30

arr

```
while(fr<=lr && fc<=lc){
    for(int j=fc; j<=lc; j++)
        sout(arr[fr][j]);
    fr++
    for(int i=fr; i<=lr; i++)
        sout(arr[i][lc];
    lc--
    for(int j=lc; j>=fc; j--)
        sout(arr[lr][j])
    lr--
    for(int i=lr; i>=fr; i--)
        sout(arr[i][fc])
    fc++
}
```

Ques: Multiply Matrices

$$
\underset{a}{\begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 1 & 2 \\ 1 & 2 & 1 \end{array}}
\times
\underset{b}{\begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 2 & 0 \\ 1 & -1 & 3 \end{array}}
=
\underset{c}{\begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 0 & 6 \\ 1 & 3 & 3 \end{array}}
$$

$c_{ij} = i^{th}$ row of $a \times j^{th}$ col of b

$$c_{00} = 1 \times 2 + 2 \times -1 = 2 - 2 = 0$$

$$c_{01} = 1 \times 0 + 2 \times 3 = 6$$

$$c_{10} = 2 \times 2 + 1 \times -1 = 3$$

$$c_{11} = 2 \times 0 + 1 \times 3 = 3$$

Ques: Multiply Matrices

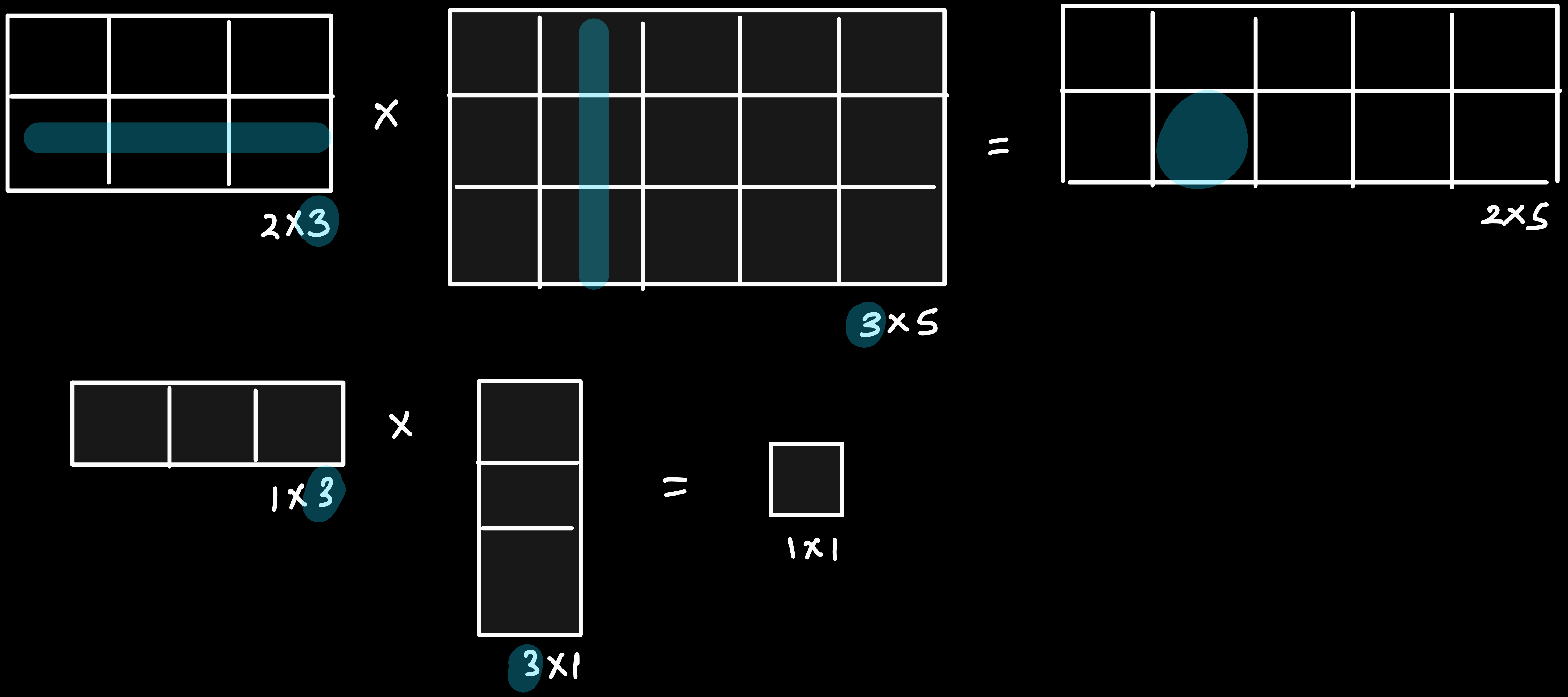
$$\begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 1 & 2 \\ 1 & 2 & 1 \end{array} \times \begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 2 & 0 \\ 1 & -1 & 3 \end{array} = \begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 0 & 6 \\ 1 & 3 & 3 \end{array}$$

a   b   c

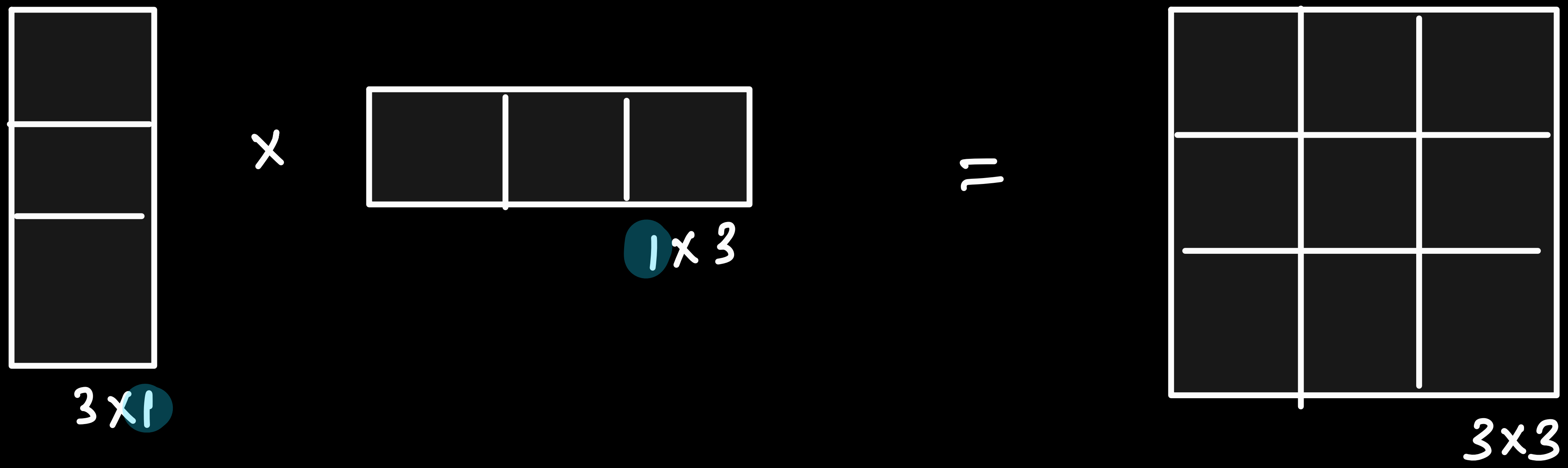
$$c[0][1] = a[0][0] * b[0][1] + a[0][1] * b[1][1]$$

$$c[i][j] = \sum_{k=0}^{n-1} a[i][k] * b[k][j]$$

Ques: Multiply Matrices



Ques: Multiply Matrices



THANKYOU

Cuties